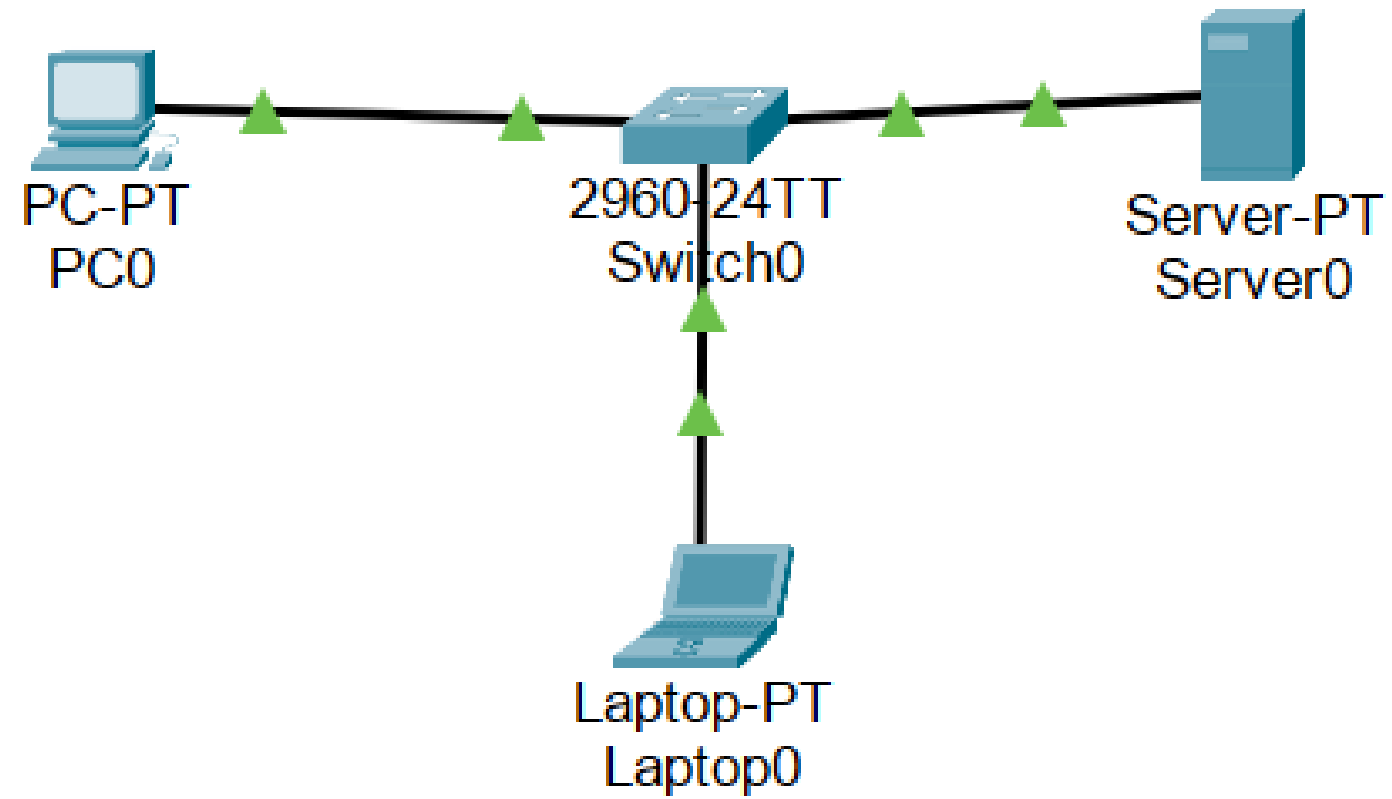


Computer Network:

A computer network is a group of interconnected devices that can communicate and share resources.

Or

A computer network is a group of interconnected devices such as computers, servers, switches, and routers that communicate with each other to share data, resources, and services using communication protocols over wired or wireless media.



Computer Networking :

Computer Networking is the process of connecting devices and enabling communication between them using protocols and network technologies.

Or

Computer networking is the process of designing, implementing, configuring, and managing networks to enable communication and data exchange between connected devices.

Internetwork :

An internetwork is a collection of multiple networks that are connected together using routers so they can communicate with each other.

In simple terms, when two or more separate networks are connected, the resulting network is called an internetwork.

Internet :

The Internet is a global internetwork, meaning it is a worldwide system of interconnected networks that communicate with each other using the TCP/IP protocol suite.

Why Do We Create a Network?

We create a computer network to allow devices to communicate, share resources, and exchange data efficiently.

Interview Definition

A network is created to enable communication between devices and to share resources such as files, printers, applications, and internet connectivity.

Main Reasons for Creating a Network

1. Resource Sharing

Networks allow multiple users to share resources.

Examples:

- Printers
- Storage
- Applications
- Internet connection

Example: In an office, 20 computers can share **one printer** instead of buying 20 printers.

2. Data Sharing

Users can easily **transfer files and information** between systems.

Examples:

- File servers
- Shared folders
- Database access

Example: Employees accessing files from a **central server**.

3. Communication

Networks enable communication between users and systems.

Examples:

- Email
- Video conferencing
- Messaging
- VoIP calls

How to Create a Computer Network ?

1. Identify the Devices (End Devices)

First, determine the devices that will be part of the network.

Examples:

- Computers (PCs / Laptops)
- Servers
- Printers
- IP Phones

These are called **end devices** or **hosts**.

2. Use Networking Devices (Intermediary Devices)

Networking devices are used to connect and control communication between hosts.

Common devices:

- **Switch** → Connects devices within the same LAN
- **Router** → Connects different networks and provides internet access
- **Access Point** → Provides wireless connectivity
- **Firewall** → Secures the network

Example structure:

PC1

PC2 → Switch → Router → Internet

PC3

3. Connect Devices Using Transmission Media

Devices must be physically or wirelessly connected.

Wired Connection

- Ethernet (UTP cables)
- Fiber optic cables

Example:

PC → Ethernet Cable → Switch

Wireless Connection

- Wi-Fi using an Access Point or Wireless Router

4. Assign IP Addresses

Every device in a network must have a **unique IP address**.

Example:

Device	IP Address
PC1	192.168.1.10
PC2	192.168.1.11
PC3	192.168.1.12
Router	192.168.1.1

This allows devices to **identify and communicate with each other**.

5. Use Networking Protocols

Communication happens using **network protocols**.

Important protocols:

- **IP** – Addressing
- **TCP / UDP** – Data transport
- **DNS** – Name resolution
- **DHCP** – Automatic IP assignment
- **HTTP / HTTPS** – Web communication

Most networks use the **TCP/IP protocol suite**.

6. Test Network Communication

After configuration, test connectivity using commands like:

Ping

Example:

```
ping 192.168.1.11
```

If replies are received, the network is working.

Simple Example (Small Office Network)

Devices required:

- 5 PCs
- 1 Switch
- 1 Router
- Ethernet cables

Network structure:

PC → Switch → Router → Internet

All PCs get IP addresses in the same network, for example:

192.168.1.0 /24

Now all devices can **communicate and share resources**.

Types of Computer Networks (CCNA Level)

Computer networks are classified based on the **geographical area they cover**.

Interview Definition

Types of networks are categories of computer networks classified according to the size or geographic area they cover.

1. PAN (Personal Area Network)

A **PAN** is a small network used to connect devices around **one person**.

Range

Usually within **10 meters**.

Examples

- Bluetooth connection
- Smartphone connected to earbuds
- Laptop connected to mobile hotspot

Technologies

- Bluetooth
- USB
- Infrared

PERSONAL AREA NETWORK



2. LAN (Local Area Network)

A **LAN** is a network that covers a **small geographical area**, such as a **home, office, or school**.

Characteristics

- High speed
- Private network
- Usually managed by one organization

Examples

- Office network
- Computer lab network
- Home Wi-Fi network

3. MAN (Metropolitan Area Network)

A MAN connects multiple LANs within a city or metropolitan area.

Examples

- City-wide internet network
- Cable TV network
- Large university campus network

Technologies

- Fiber optic networks
- Metro Ethernet

4. WAN (Wide Area Network)

A **WAN** covers a large geographical area, such as countries or continents.

It connects multiple LANs and MANs together.

Examples

- Bank branch networks
- Company branch office networks
- ISP networks

Technologies

- MPLS
- Leased lines
- Satellite links
- Fiber backbone

Quick Summary

Network Type	Coverage Area
PAN	Personal devices (few meters)
LAN	Home, office, school
MAN	City-level network
WAN	Country or global network

Best CCNA Interview Answer

Computer networks are categorized based on geographical coverage into PAN, LAN, MAN, and WAN. PAN connects personal devices, LAN covers small areas like offices, MAN covers a city, and WAN connects networks across large geographical areas such as countries or continents.

Network Architecture

Network architecture refers to the overall design and structure of a computer network, including how devices are connected, how data flows, and which protocols and technologies are used.

Interview Definition

Network architecture is the logical and physical design of a network that defines how devices communicate, how data is transmitted, and how network resources are organized.

Main Types of Network Architecture

1. Client–Server Architecture
2. Peer-to-Peer (P2P) Architecture

1. Client–Server Architecture

In **client-server architecture**, one or more **servers provide services** and multiple **clients request those services**.

How it Works

- **Server** stores resources and manages the network.
- **Clients** request services from the server.

Example:

- Web server hosting a website
- File server storing company documents
- Email server handling emails

Example network:

Client PCs → Switch → Server

Advantages

- Centralized management
- Better security control
- Easier backup and maintenance

Disadvantages

- Server failure can affect clients
- Higher setup cost

2. Peer-to-Peer (P2P) Architecture

In **peer-to-peer architecture**, all devices are **equal**, and each device can act as both a **client** and a **server**.

How it Works

Devices share resources directly with each other without a central server.

Example:

- File sharing between two computers
- Small home networks

Example network:

PC1 ↔ PC2 ↔ PC3

Advantages

- Easy to set up
- Low cost
- No dedicated server required

Disadvantages

- Poor security
- Difficult management
- Not suitable for large networks

Comparison

Feature	Client-Server	Peer-to-Peer
Central server	Yes	No
Security	High	Low
Management	Centralized	Decentralized
Suitable for	Large organizations	Small networks
Cost	Higher	Lower

Network Topology (CCNA Level)

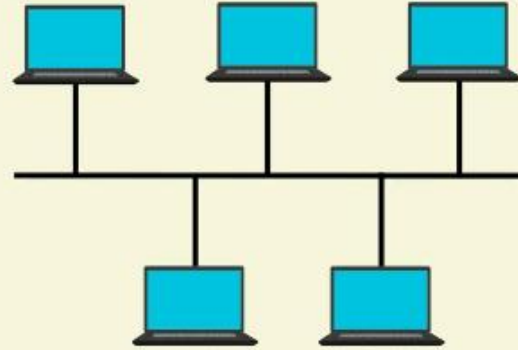
Network topology refers to the **physical or logical arrangement of devices in a network**, showing how devices are connected and how data flows between them.

Interview Definition

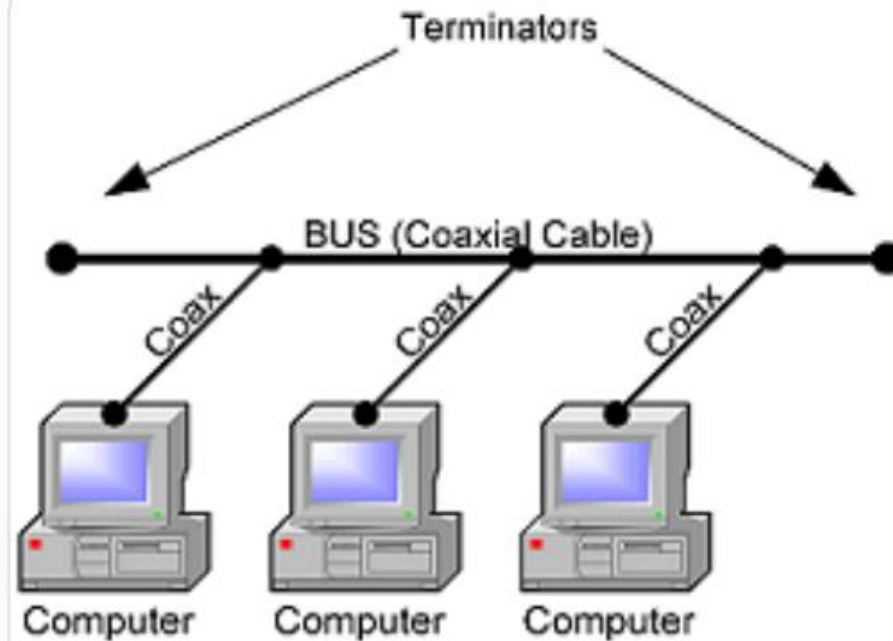
Network topology is the layout or structure of a network that defines how devices such as computers, switches, and routers are interconnected.

Types of Network Topology

1. Bus Topology



BUS TOPOLOGY



In **bus topology**, all devices are connected to a **single central cable** called a **backbone**.

How it Works

- Data travels along the **main cable**
- All devices receive the signal
- Only the intended device accepts the data

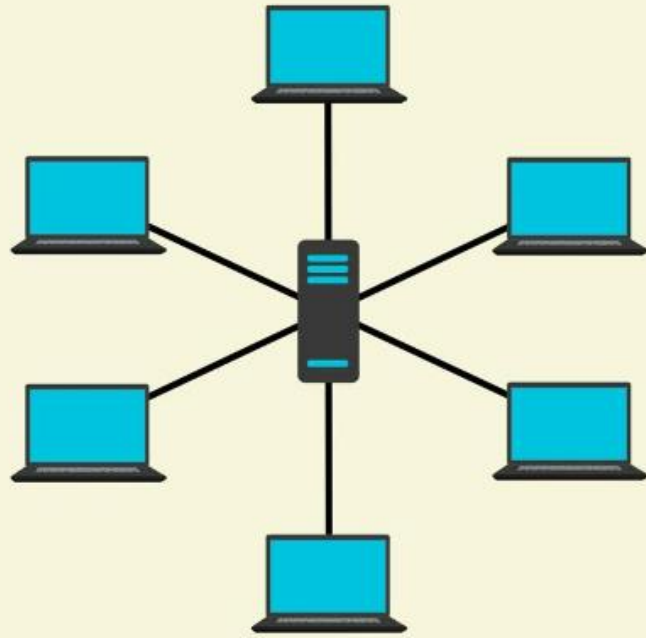
Advantages

- Simple design
- Low cost
- Uses less cable

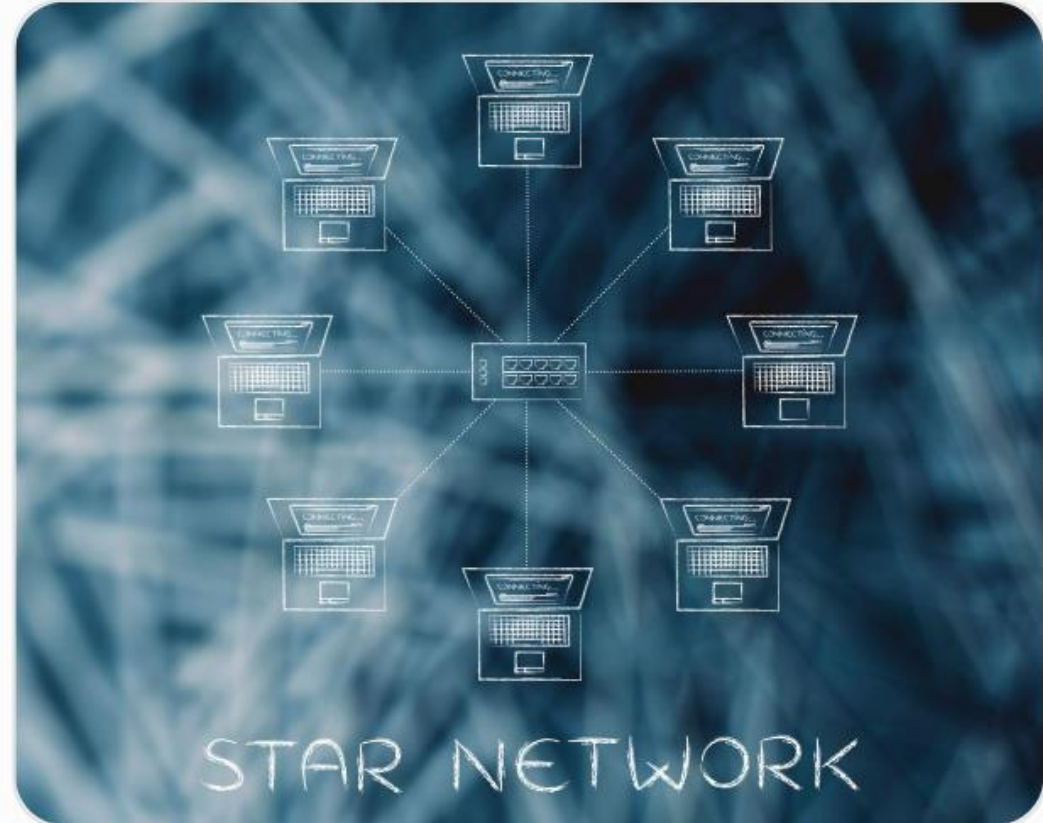
Disadvantages

- If backbone cable fails → entire network fails
- Difficult troubleshooting
- Limited scalability

2. Star Topology



STAR TOPOLOGY



In star topology, all devices connect to a **central device** like a switch or hub.

How it Works

Each device has a **separate cable connection** to the central device.

Example:

PC → Switch

PC → Switch

PC → Switch

Advantages

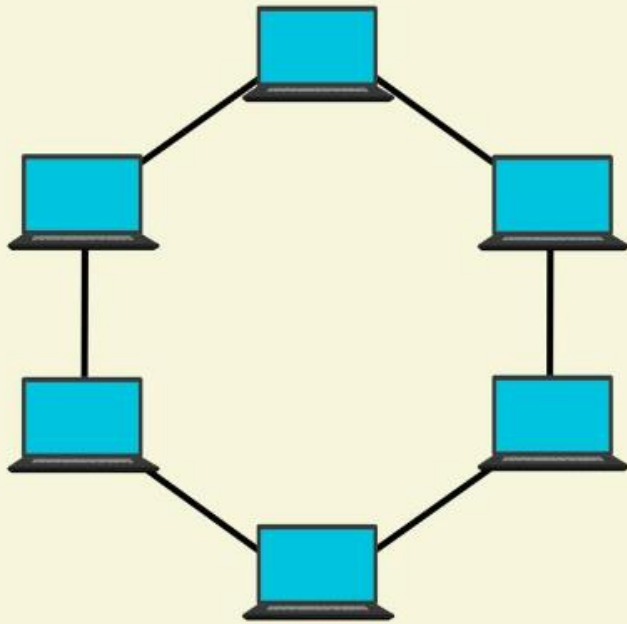
- Easy to manage
- Easy to troubleshoot
- One cable failure affects only one device

Disadvantages

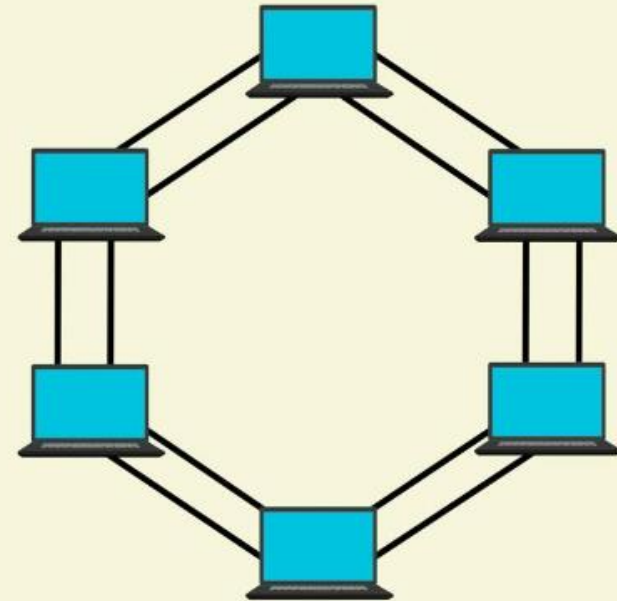
- If central device fails → entire network fails
- Requires more cable

Most modern LAN networks use star topology.

3. Ring Topology



RING TOPOLOGY



DOUBLE RING TOPOLOGY

In ring topology, devices are connected in a **circular loop**.

How it Works

Data travels in **one direction around the ring** until it reaches the destination.

Example:

PC1 → PC2 → PC3 → PC4 → PC1

Advantages

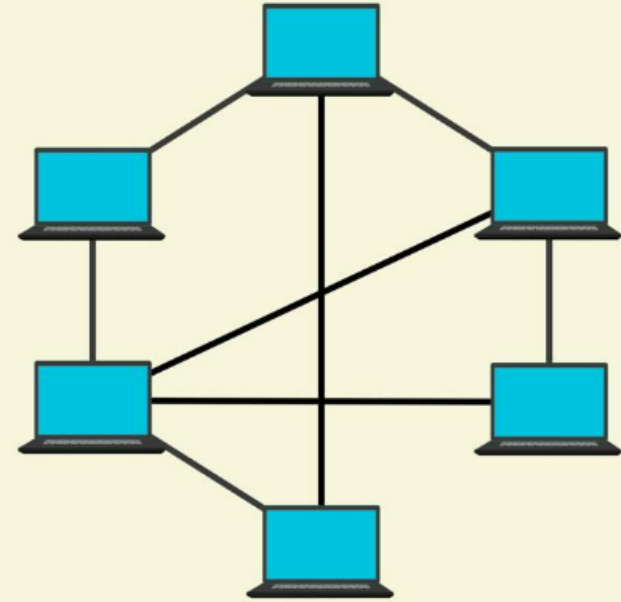
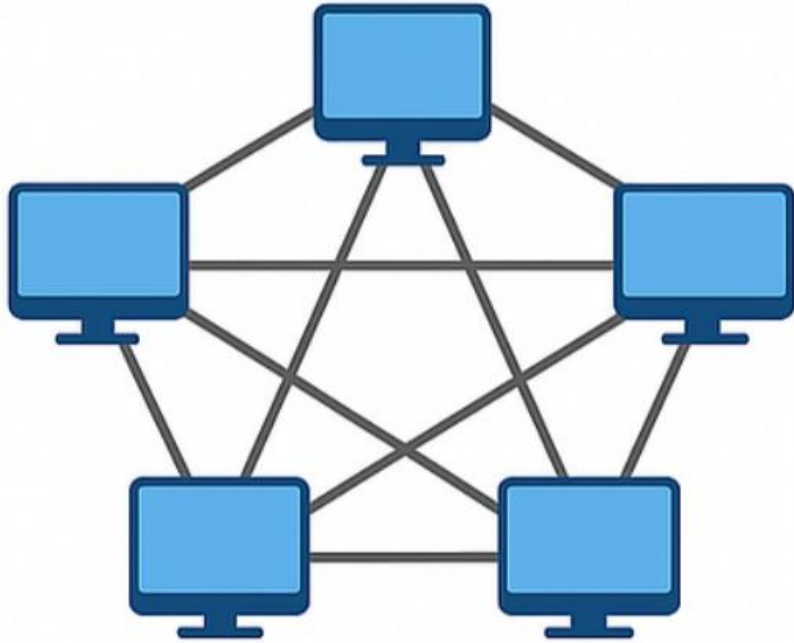
- No data collision
- Predictable performance

Disadvantages

- If one device fails → entire network can fail
- Difficult troubleshooting

4. Mesh Topology

Mesh Topology



M E S H T O P O L O G Y

In mesh topology, every device connects to **multiple other devices**.

Types

- **Full Mesh** – every device connected to every other device
- **Partial Mesh** – some devices interconnected

Advantages

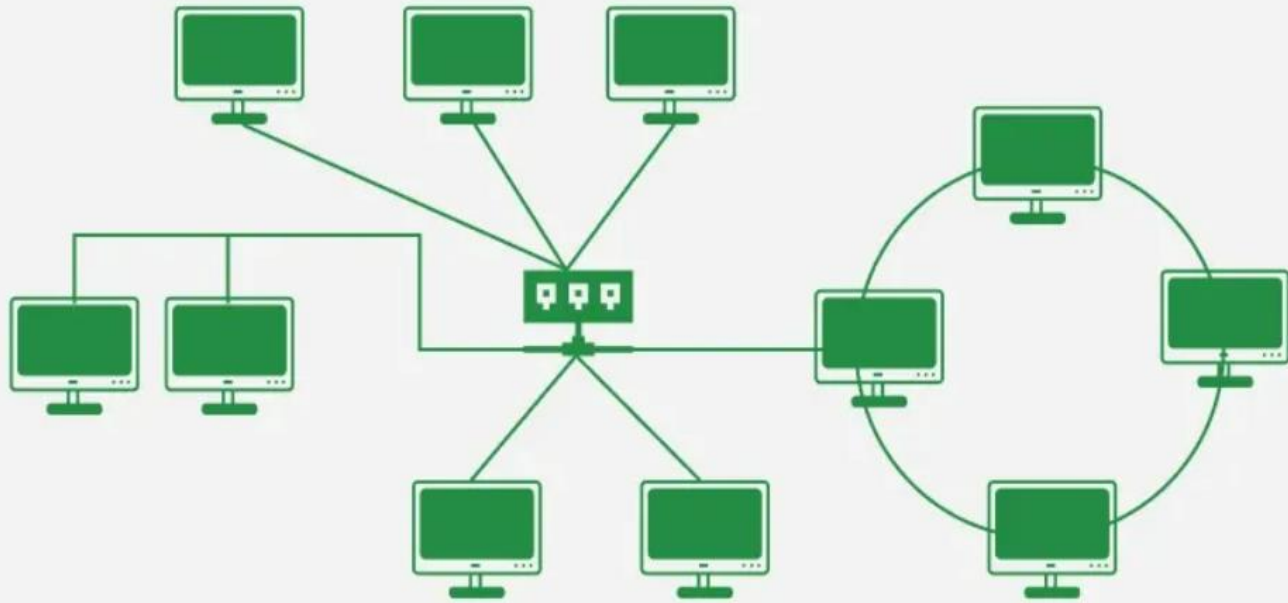
- High reliability
- Multiple paths for data
- No single point of failure

Disadvantages

- Expensive
- Complex configuration

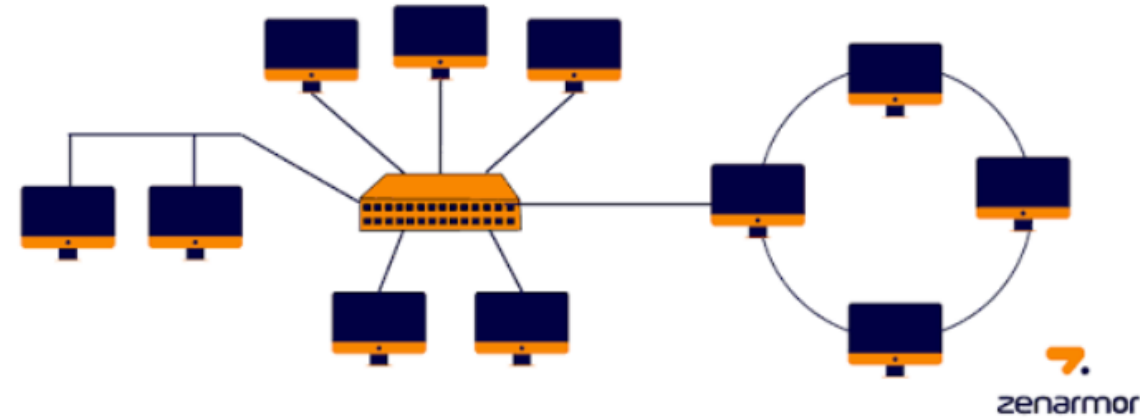
Used in **WAN** and **backbone networks**.

5. Hybrid Topology



HYBRID NETWORK TOPOLOGY

A hybrid topology is a type of network topology that combines two or more network topologies, including ring, bus, and mesh topologies



A hybrid topology combines two or more topologies.

Example:

- Star + Bus
- Star + Mesh

Most modern enterprise networks use hybrid topology.

Quick Summary

Topology	Structure
Bus	Single backbone cable
Star	All devices connect to central switch
Ring	Devices connected in a circular loop
Mesh	Devices interconnected with multiple paths
Hybrid	Combination of multiple topologies

Best CCNA Interview Answer

Network topology refers to the physical or logical layout of a network that defines how devices are connected and how data flows between them. Common types include bus, star, ring, mesh, and hybrid topology.

✓ Data Transmission Modes (Direction of Data Flow)

1 Simplex Mode (One-Way Communication)



📌 Explanation

In **simplex mode**, data flows in **only one direction**.

One device is always the sender, and the other is always the receiver.

Key Points

- No reverse communication
- No acknowledgment
- Used where feedback is not required

Examples

- Keyboard → Computer
- TV broadcasting

Interview Line

“Simplex mode allows communication in only one direction, where one device acts as sender and the other as receiver.”

2 Half-Duplex Mode (Two-Way, Not Simultaneous)



Explanation

In half-duplex mode, both devices can send and receive data, but **not at the same time**.



Key Points

- Communication is bidirectional
- Devices take turns
- Requires coordination



Examples

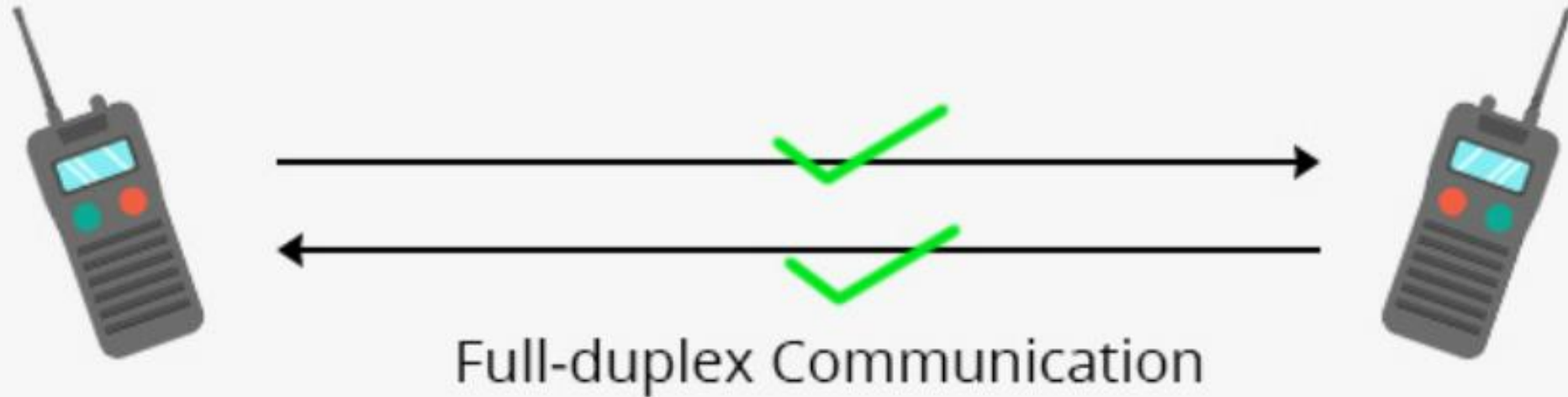
- Walkie-talkie
- CB radio



Interview Line

“Half-duplex mode allows communication in both directions, but only one device can transmit at a time.”

3 Full-Duplex Mode (Two-Way Simultaneous)



Explanation

In full-duplex mode, both devices can send and receive data at the same time.



Key Points

- No waiting or switching
- High efficiency
- No collisions in Ethernet



Examples

- Phone calls
- Video conferencing
- Modern LAN (switch-based Ethernet)



Interview Line

“Full-duplex mode enables simultaneous two-way communication, making it the most efficient transmission mode.”

Quick Comparison (Must Remember)

Feature	Simplex	Half Duplex	Full Duplex
Direction	One-way	Two-way (one at a time)	Two-way (simultaneous)
Communication	No feedback	Turn-based	Real-time both sides
Efficiency	Low	Medium	High
Example	Keyboard	Walkie-talkie	Phone call

Important Interview Twist (Very Important)

 Interviewer may confuse you with:

- Transmission Mode vs Transmission Type

 Transmission Mode = Direction

- Simplex / Half / Full Duplex

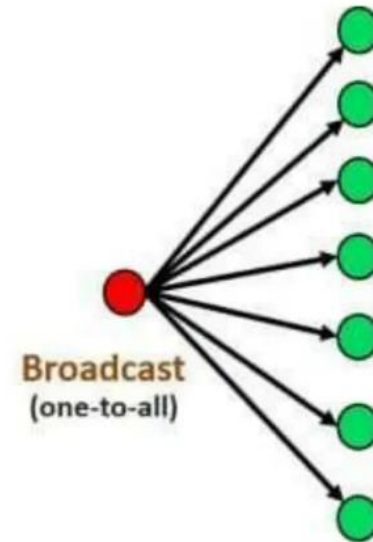
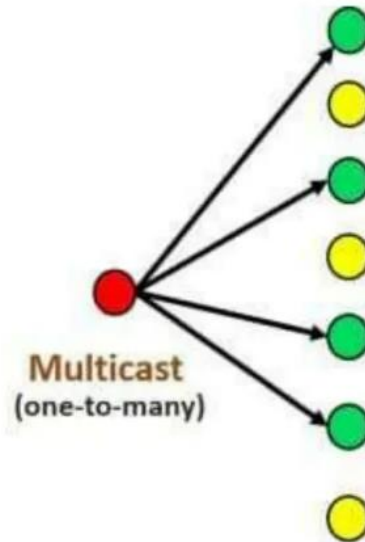
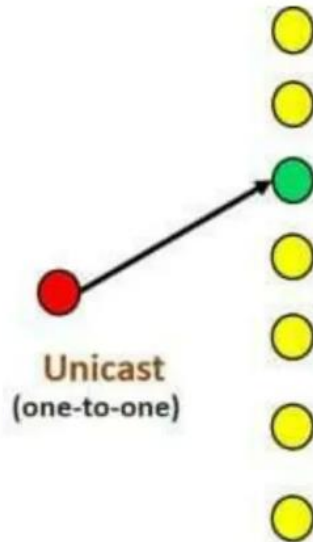
 Transmission Type = How data travels

- Serial / Parallel

✓ Types of Casting (Data Delivery Methods)

There are 3 main types you must know:

1 Unicast (One-to-One Communication)



📌 Definition

Unicast is communication between one sender and one specific receiver.

Key Points

- Data goes to **only one device**
- Uses **unique IP address**
- Most common type in networks

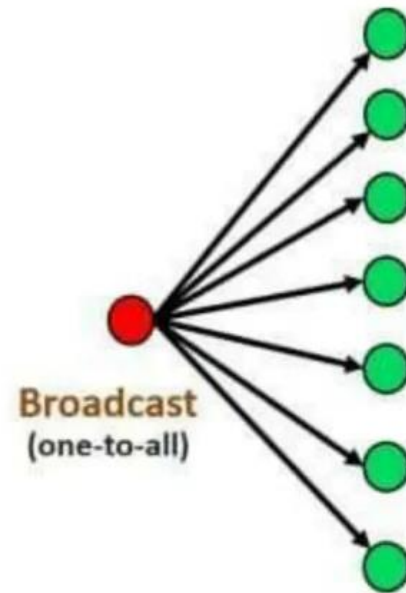
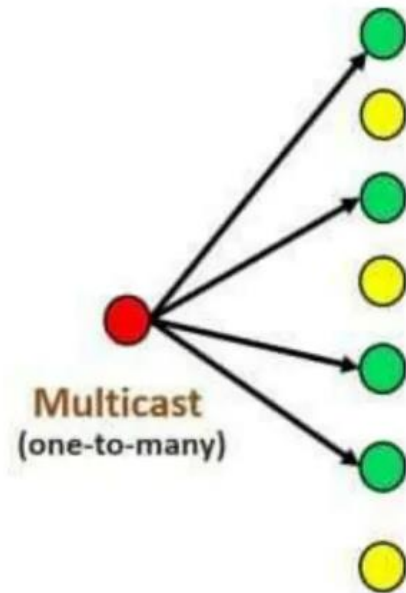
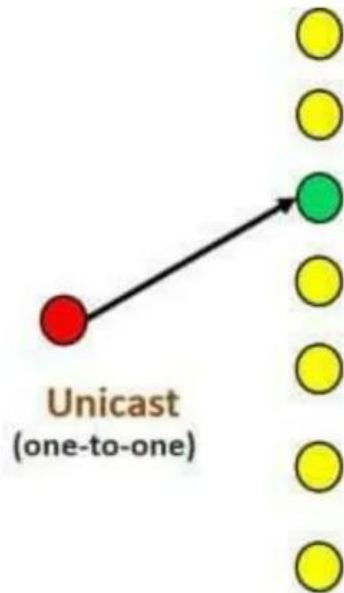
Examples

- Opening a website
- Sending an email
- SSH connection

Interview Line

“Unicast is a one-to-one communication where data is sent from one sender to a specific receiver using a unique IP address.”

2 Broadcast (One-to-All Communication)



Definition

Broadcast is communication where data is sent from one sender to all devices in the network.

Key Points

- Sent to **all devices in same network**
- Uses **broadcast IP address**
- Can cause network traffic if overused

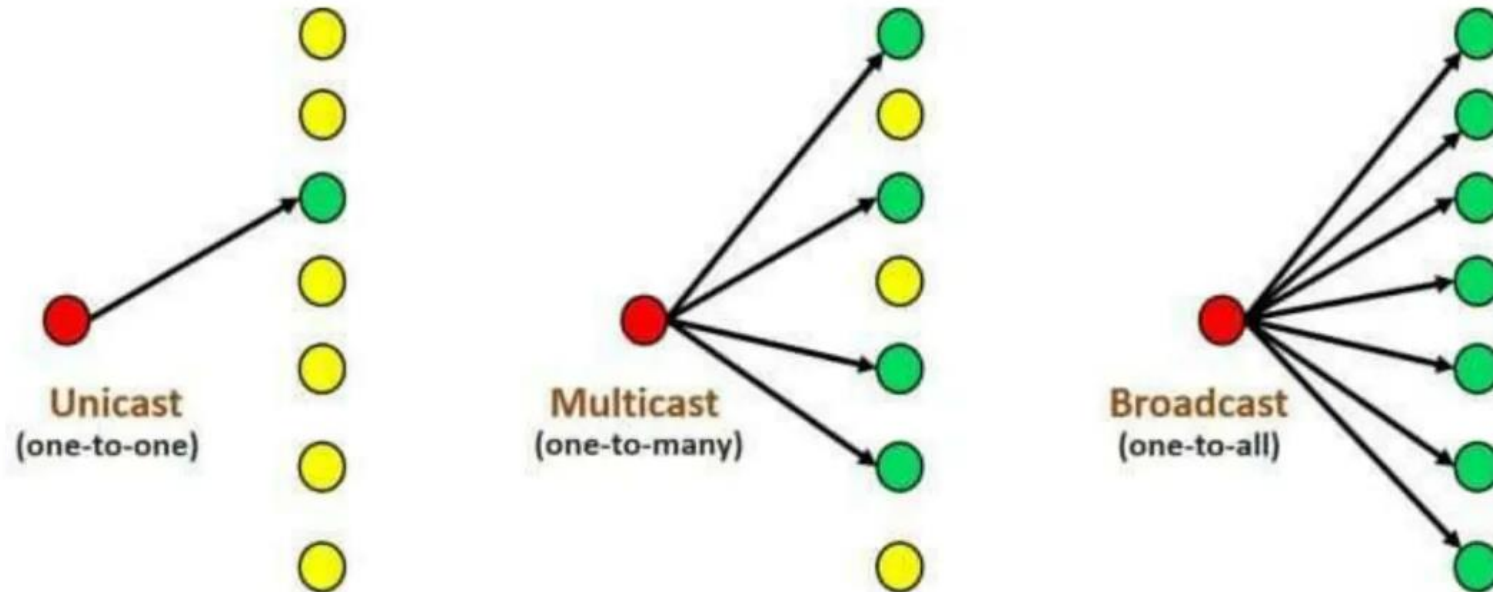
Examples

- ARP request
- DHCP discover

Interview Line

“Broadcast sends data from one sender to all devices in the network using a broadcast address.”

3 Multicast (One-to-Many Selected)



Definition

Multicast is communication where data is sent from one sender to a specific group of receivers.

Key Points

- Not all devices, only **selected group**
- Efficient use of bandwidth
- Uses **multicast IP range (224.0.0.0 – 239.255.255.255)**

Examples

- Live video streaming
- Online gaming updates
- IPTV

Interview Line

“Multicast is a one-to-many communication where data is sent to a specific group of devices instead of all devices.”



Quick Comparison (Important for Interview)

Type	Communication	Target Devices	Example
Unicast	One-to-One	Single device	Web browsing
Broadcast	One-to-All	All devices in LAN	ARP request
Multicast	One-to-Many	Selected group	Video streaming



Extra (If Interviewer Goes Deep)

? What about Anycast?

👉 Advanced concept (not basic CCNA)

- **Anycast = One-to-Nearest**
- Data goes to **closest server**
- Used in DNS, CDNs